

Client Statistical Analysis Report

Client: Portfolio Demo / Custom CSV Project | **Prepared by:** Tianyou Liu - Statistical Analysis | **Date:** June 18, 2026

Executive Summary

High-level decision summary for business stakeholders before the full statistical methodology.

TEST	Welch's T-Test (Unequal Variances)
RESULT	A statistically significant difference was detected. $p = 5.21e-05$ at $\alpha = 0.05$.
RECOMMENDATION	Use Variant if a higher metric value is better for the business objective.
CONFIDENCE	95% CI for Group B - Group A: [4.9964, 13.8078]
PLAIN ENGLISH	The Welch's T-Test (Unequal Variances) found a statistically significant result ($p = 5.21e-05$, $\alpha = 0.05$). Variant was higher than Control in this sample.

1. State

Define the hypotheses and significance level to evaluate the claims of the A/B test.

Significance Level (α): 0.05

Hypotheses (Symbols):

$$H_0 : \mu_A = \mu_B \quad H_a : \mu_A \neq \mu_B$$

Null Hypothesis (H_0): There is no difference in the means of the metrics between Control and Variant ($\mu_A = \mu_B$).

Alternative Hypothesis (H_a): There is a statistically significant difference in the means of the metrics between Control and Variant ($\mu_A \neq \mu_B$).

2. Plan

Identify the appropriate statistical procedure and verify its mathematical assumptions.

Recommended Procedure: Welch's T-Test (Unequal Variances)

Assumption / Condition Checked	Status & Diagnostic Details
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Randomness	Assuming independent random sampling/assignment of groups.
Independence	Samples are mutually independent and represent < 10% of population.
Normality (Shapiro-Wilk)	Group A Normality (p = 0.7219, Passed). Group B Normality (p = 0.8531, Passed). Normality met.
Equal Variance (Levene's)	Levene's Test for equal variance: p = 0.0150 (Unequal variances assumed).

CONDITIONS MET

All mathematical conditions for the Welch's T-Test (Unequal Variances) are satisfied. Parametric results are highly reliable.

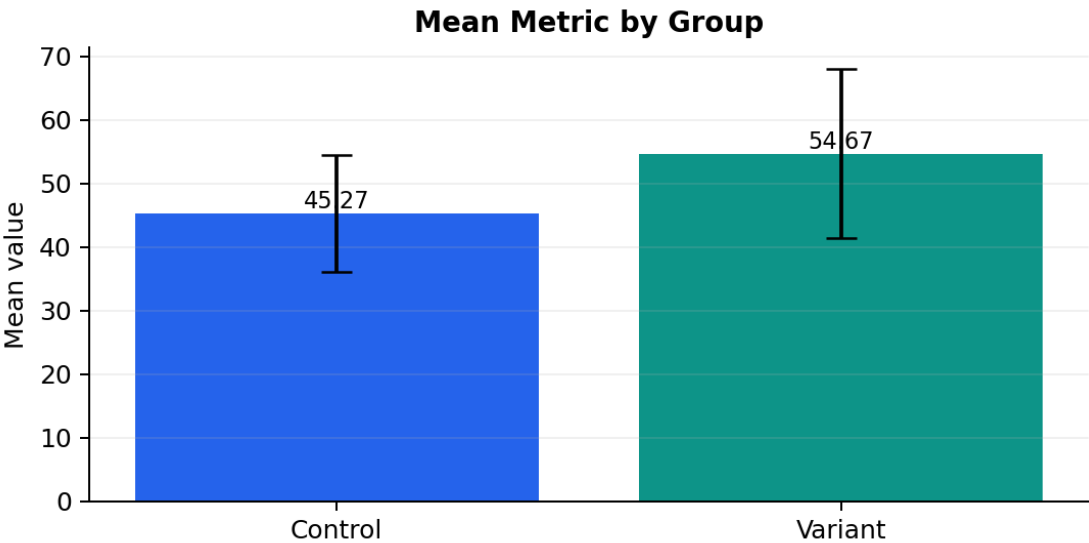
3. Do

Calculate sample metrics, standard error, test statistic, and the resulting p-value.

Sample Descriptive Statistics:

Group	Sample Size (n)	Sample Mean ($\bar{x}$)	Standard Deviation (s)	Variance (s^2)
Control	50	45.2701	9.2023	84.6828
Variant	55	54.6722	13.3346	177.8121

Visual Summary:



Calculations & Mathematical Formulations:

- Se:

$$SE = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}} = 2.2196$$

• *T Stat:*

$$t = \frac{\bar{x}_B - \bar{x}_A}{SE} = 4.2360$$

• *Df:*

$$df = 96.28$$

• *CI:*

$$CI = (\bar{x}_B - \bar{x}_A) \pm t_{crit} \times SE = (4.9964, 13.8078)$$

Statistical Parameter	Value
Test Statistic (t)	4.23595
p-value	5.21118e-05
Degrees of Freedom (df)	96.28
95% Confidence Interval	(4.9964, 13.8078)
Cohen's d	0.8137 (Standardized mean difference: Group B - Group A)

4. Conclude

State the final decision and offer a business interpretation based on the statistical results.

Compare p-value to α : $0.00005 < 0.05$. Since the p-value is less than the significance level, we **reject the null hypothesis (H_0)**. There is strong evidence supporting a statistically significant difference between the two variations.

DECISION: REJECT NULL HYPOTHESIS

Business Action Recommendation: The metric difference of 9.4021 is statistically significant. Variation **Group B** outperforms the other and should be adopted to maximize efficiency/revenue.

Plain English:

The Welch's T-Test (Unequal Variances) found a statistically significant result ($p = 5.21e-05$, $\alpha = 0.05$). Variant was higher than Control in this sample.